

Quantum Mechanics without Ontology?

Configuration Space, the Arena Problem, and the Road Not Taken in 1926

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Abstract. It is often said that quantum mechanics dispenses with ontology and that its predictive success licenses the omission. This paper argues that the situation is both more precise and more serious. Under the preparation-independence assumption, the Pusey–Barrett–Rudolph theorem tells against a purely epistemic reading of the quantum state, so the state is naturally taken to be ontic; the genuine difficulty is not the existence of an ontology but its *arena* — the state is a function on $3N$ -dimensional configuration space, not on the three-dimensional space that observers and instruments occupy. The measurement problem, the status of collapse, and the proliferation of interpretations are read as symptoms of this arena problem. The paper revisits the historical moment (spring 1926) at which the wave function was lifted from three-dimensional space to configuration space over the objections of Schrödinger and Einstein; situates the options within the contemporary debate between wave-function realism and primitive ontology; presents a three-dimensional charge-density formulation as an under-explored primitive-ontology alternative — a concrete realization of the road not taken; and draws a practical corollary from quantum computing, whose claimed advantage turns on the physical reality of the exponentially large amplitude.

Keywords. quantum state; ontology; configuration space; wave-function realism; primitive ontology; PBR theorem; measurement problem; history of quantum mechanics.